HEA

UUM
Universiti Utara Malaysia

FINAL EXAMINATION
SECOND SEMESTER 2023/2024 SESSION

COURSE CODE/NAME	:	STIW3064 / COMPONENT-BASED DEVELOPMENT
DATE	:	28 JULY 2024 (SUNDAY)
TIME	:	9.00 A.M. – 11.30 A.M. (2 ½ HOURS)
VENUE	:	DP1 – DP2 STHEM

INSTRUCTIONS :

1. This examination paper has **ELEVEN (11)** questions in **EIGHT (8)** printed pages excluding the cover page.
2. Answer **ALL** questions in the space provided.
3. You are **NOT ALLOWED** to remove the examination paper from the examination hall.

MATRIC NO.: _____ (in words) | | | | | | | (in numbers)

IDENTIFICATION CARD NO.: | | | | | | | | | | | | |

LECTURER NAME: _____

GROUP: | | **TABLE NO.:** | | | | |

DO NOT OPEN THIS EXAMINATION PAPER UNTIL INSTRUCTED

INSTRUCTION: ANSWER ALL QUESTIONS IN THE SPACE PROVIDED.

1. Software development life-cycle (SDLC) and waterfall models are often mistakenly being used interchangeably.

   a) Differentiate both models by highlighting **ONE (1)** of their major distinguishing characteristics.

   (4 marks)

   b) Explain **ONE (1)** main reason why both models are always being interchangeably used.

   (4 marks)

   c) State the category of software development approach that the waterfall model falls into.

   (2 marks)

2. One of the approaches in software development is component-based development (CBD).

   a) Describe the main concept that underlies the CBD approach.

   (4 marks)

   b) Explain **ONE (1)** situation in which component-based development is feasible.

   (4 marks)

   c) State **TWO (2)** main outcomes of component-based development.

   (2 marks)

3. Similar to other conventional software development approaches, component-based development (CBD) also follows the processes in the software development life cycle. However, several activities in CBD processes differ significantly from the conventional development processes, and this mainly relates to the use and reuse of components.

   a) Describe **TWO (2)** activities that are unique to component-based development.

   (10 marks)

   b) For any **ONE (1)** of the activities stated in **3(a)**, justify its corresponding activity in the waterfall model.

   (5 marks)

4. In component-based development, priority is given to the reuse of readily available components. However, reusable components might not be available all the time.

   a) Determine what needs to be done by the developer when reusable components are not available for reuse.

   (2 marks)

   b) Identify **TWO (2)** models that could be applied in performing the activity in **4(a)**.

   (2 marks)

   c) State the desired outcome of the process or activity performed in **4(a).**

   (2 marks)

5. Discuss **ONE (1)** of the activities that need to be performed before the component development process takes place.

   (5 marks)

6. In component-based development, the selection of the best-fit components must be appropriately performed and this involves several activities.

   a) Analyze how the involved activities are being performed in the selection process.
   (12 marks)

   b) Determine the main outcomes of the activities mentioned in **6(a)**.
   (3 marks)

7. A component model is used both in component development and component-based development, but it is meant to meet different purposes. Differentiate between the two models based on their purposes.

(10 marks)

8. There are issues in deploying components that were developed for certain architectures. Justify **ONE (1)** of the issues that a developer might face to deploy a component on a specific architecture.

(5 marks)

9. The definition of a software component is given below:

> ***Software Component Definition:***
>
> *"A component is a software element that conforms to a component model and can be independently deployed and composed without modification according to a composition standard."*

For a component that will undergo the composition and integration process, justify **TWO (2)** impacts of the given definition on the component.

(10 marks)

10. A system that is developed using component-based development (CBD) will undergo similar testing processes as systems that are developed using other methods. State **ALL** types of testing that are pertinent to the system that is developed using CBD.

(4 marks)

11. Components that have been developed shall be securely placed in repositories to allow them to be reused in other development projects. There are two repositories that are commonly used to store the components namely the Testable Components Repository (TCR) and the Reusable Component Repository (RCR).

    a) Differentiate between the TCR and RCR.

(5 marks)

    b) How could a reusable component be retrieved from the RCR?

(5 marks)

END OF QUESTIONS